MTH320

Specimen Examination Paper

Mathematical Methods II

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **THREE (3)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **open-book** examination.

At the end of the examination
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

Answer all questions. (Total 100 marks)

Question 1

A damped mass-spring has the following equation of motion with initial conditions:

$$y'' + 2y' + y = 10, \quad y(0) = 0,\ y'(0) = 25$$

in which y(t) represents the displacement function of time.

(a) Formulate a system of 1st-order ODEs from the given initial value problem.

(5 marks)

(b) Using the Duhamel's principle, solve the system of ODEs obtained in Question 1(a). Hence, determine the displacement function y(t) of the mass.

(20 marks)

Question 2

Figure Q2 shows a pendulum of mass m being hinged at a fixed point O and undergoes a swinging motion. The mass is subjected to an external force f(t) that is always tangential to the rod of length L. The mass of the rod is negligible and the hinge can be treated as frictionless. The air resistance is also insignificant.

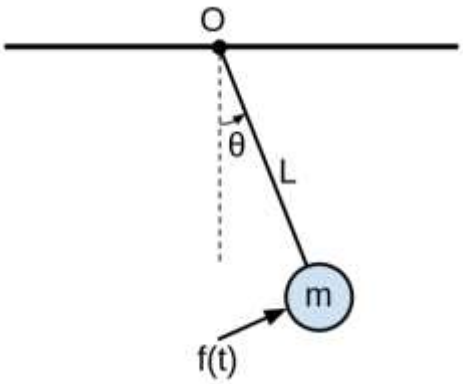

Figure Q2: Pendulum swinging with angular position function, θ(t)

(a) Sketch a free-body diagram of the pendulum mass with all forces clearly labelled. By applying Newton's 2nd law on the mass and considering small angular displacements, show that the governing equation for the angular motion of the pendulum is:

$$\theta'' + \frac{g}{L}\theta = \frac{f(t)}{mL}$$

where g represents the acceleration due to gravity.

(6 marks)

(b) The tangential force f(t) has the expression:

$$f(t) = Ae^{-t} + B\cos(\beta t), \quad \beta = \sqrt{\frac{g}{L}}$$

where A and B are constants. Using the method of undetermined coefficients, solve the equation of motion in Question 2(a) for the general solution of the angular position function θ(t).

(16 marks)

Question 3

(a) Determine the Laplace transform of:

$$f(t) = 2t + tu(t-1).$$

(5 marks)

(b) Hence, solve the following IVP by Laplace transform.

$$2y - 3y' = f(t) - 2t, \; y(0) = 1$$

(15 marks)

Question 4

Solve the heat conduction problem along a 3-meter rod described by:

$$\begin{aligned} \alpha T_{xx} &= T_t && , \; 0 < x < 3, \; t > 0 \\ T(0,t) = 300, \;\; T(3,t) &= 0 && , \; t > 0 \\ T(x,0) &= 30\cos(2\pi x)\sin(2\pi x) + 100(3-x), && 0 < x < 3 \end{aligned}$$

(16 marks)

Question 5

Solve the Laplace equation problem on a 2 by 3 meter rectangle described by:

$$\begin{aligned} u_{xx} + u_{yy} &= 0 && , \; 0 < x < 2, \; 0 < y < 3 \\ u(0,y) = 0, \;\; u(2,y) &= -1, && 0 < y < 3 \\ u(x,0) = 0, \;\; u(x,3) &= 1 \; , && 0 < x < 2 \end{aligned}$$

(17 marks)

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